

2 Goal and Area of Application

A general goal of drug repurposing projects is to select compounds that have been approved by regulatory agencies, or that are in advanced clinical trial phases and could be prioritized to be assayed for new indications. The *in silico* analysis of databases and the development of predictive tools offers an opportunity to increase the efficiency of this process.

This SOP describes a potential approach to use *in silico* tools and databases to establish a drug repurposing workflow with the goal of identifying drugs for new indications, analyze their possible mechanisms of action (targets and pathways), and evaluate other properties like pharmacokinetics and safety.

3 Definitions and Abbreviations

MoA	Mechanism of Action
SRS	Spontaneous Reporting Systems
FAERS	FDA Adverse Event Reporting System
VAERS	Vaccine Adverse Event Reporting System
JADER	Japanese Adverse Event Drug Report
FDA	Food and Drug Administration
PMDA	Pharmaceuticals and Medical Devices Agency (Japan)

4 Method

4.1 General

Numerous and diverse *in silico* tools and databases for drug discovery and development have been published. Within the REMEDI4ALL project, many of these have been collected, analyzed, prioritized, and brought together on a dedicated website developed for this purpose: <https://idrc-r4a.com/landing>.

In this SOP, we use some of these *in silico* tools and databases to establish a drug discovery repurposing workflow that, starting from an indication, would:

- analyzes pathways and targets possibly associated with such an indication;
- selects compounds approved by regulatory agencies or in advanced clinical development phases, for other indications that could interact with the repurposed indication targets
- prioritizes the selected compounds based on known or predicted physicochemical characteristics and/or based on their known or predicted safety events.

This SOP presents one of the workflows explored during the 1st REMEDI4ALL Hackathon, and a full guideline used for this event is to be found as an annex .

4.2 Information on tools and resources

Resource name	Resource use	Link
Reactome	Pathways	https://reactome.org/
OpenTargets	Targets	https://platform.opentargets.org/
DISGENET	Disease-gene associations	https://www.disgenet.org/

ChEMBL	Compounds-targets interactions	https://www.ebi.ac.uk/chembl/
PubChem	Compounds-targets interactions	https://pubchem.ncbi.nlm.nih.gov/
CompTox	Chemicals-toxicity	https://comptox.epa.gov/dashboard/
ClarityVista	Safety analysis	https://clarity-vista.com/

4.3 Procedure

Researchers worldwide have been studying the interactions of compounds (including small molecules like aspirin or biologicals like vaccines, proteins, antibodies, etc.), with different components of our cells (usually proteins), which are generally called targets. They have incorporated the results of this extensive research into several databases that contain the compounds, their targets, and the diseases (indications) for which they have been studied. For repurposing projects, knowing the compound-target interactions (i.e. how a drug affects a specific component in the cell to produce the desired effect), and the target-disease associations (i.e. the connection between a specific molecule in the cell and a particular disease) is crucial to propose new uses for existing medications.

Pathway and target exploration

The first two questions to be addressed in this workflow are:

- What are the biological pathways associated with the assigned disease?
- What are the (protein) targets associated with these disease pathways?

To answer these questions, we first explored the Reactome database (<https://reactome.org/>) to extract relevant pathways for the disease of interest and identify key proteins or genes within these pathways that could be potential targets for intervention. Additionally, we explored OpenTargets and DISGENET in the following manner:

- Go to the Open Targets platform (<https://www.targetvalidation.org/>).
 - Search for your disease or condition of interest.
 - Explore the associated targets and their evidence scores. Targets with higher scores are more likely to be relevant.
 - Pay attention to the functional evidence supporting each target.
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- Access the DISGENET database (<https://www.disgenet.org/>)
 - Search for your disease.
 - Explore the genes associated with the disease and assess their relevance as potential targets. DISGENET provides information on gene-disease associations from various sources, helping you identify genes strongly linked to your disease of interest.

The exploration of these databases will give you an initial list of the targets associated with your assigned disease. Explore this initial list and combine this information with your knowledge of the

disease. You may ask yourself the question: Are the identified targets relevant to the disease or phenotype of interest?

Evaluate the relevance of the identified targets based on the disease context. Consider whether the targets align with known disease pathways, genetic associations, or molecular mechanisms and prioritize the identified targets based on their potential significance in the disease process.

- a. Combine the information gathered from Reactome, Open Targets, and DiGeNet. Prioritize potential targets based on:
 - Consistency across different databases (if a target is implicated in multiple sources, it may be more reliable).
 - Strength of evidence (e.g., targets with higher evidence scores in Open Targets).
- b. Use your judgment to prioritize targets that are most likely to yield therapeutic benefits, and end with a list of approximately ten candidate targets.

Selection of compounds interacting with targets

In this section of the workflow, we address the question: what are the (small-molecule) compounds on the market or in clinical trials associated with the selected targets, and their bioactivities?

In this step, the goal is to identify all the approved compounds (or those in advanced clinical trials), linked to your list of targets selected in section 4.2. The database proposed to be explored is the [ChEMBL](https://www.ebi.ac.uk/chembl/) database (<https://www.ebi.ac.uk/chembl/>).

The ChEMBL database^{1, 2} is a manually curated database of bioactive molecules with drug-like properties. It brings together chemical, bioactivity, and genomic data to aid the translation of genomic information into effective new drugs. It is a large-scale, open-access database of bioactive molecules, their targets, and their associated activities. It primarily focuses on small molecules and their interactions with biological macromolecules such as proteins and nucleic acids.

To select compounds that interact with the prioritized targets, using ChEMBL, the following workflow is proposed:

For each prioritized target:

- Go to the targets area of ChEMBL:
https://www.ebi.ac.uk/chembl/web_components/explore/targets/
- Search one of your prioritized targets and filter results e.g. by species to find the best-fitting entry.
- Click on that best-fitting target to see the entry, then click on “Activity Charts” in the navigation panel, and then on “Activity types for...” above the left-hand chart. You may now

¹ More information about ChEMBL can be found at: <https://chembl.gitbook.io/chembl-interface-documentation/about>

² European Bioinformatics Institute - EMBL-EBI (2021) *A guide to exploring drug like compounds and their biological targets using ChEMBL*, <https://www.youtube.com/watch?v=Yi8RIIYseS8>

explore which compounds interact with your target and their chemical properties – which characteristics may affect their suitability as drugs for the target condition?

- Now click on the Configuration icon “show / hide / organize columns” above the molecule (not the filtering) pane. Here, select Molecule Max Phase (column number 22). For ease of view, you can also drag and drop this item to the top of the list. It will appear as one of the first columns in the molecule table. The Molecular Max Phase describes the latest phase of clinical development that the molecule has reached, as indicated by a simple numerical value.
 - A score of 2 tells us the molecule has reached phase II clinical trials.
 - A score of 3 tells us a molecule has reached Phase III clinical trials.
 - A score of 4 tells us a molecule has been approved to treat a condition.
 - For each target in your list, extract the compounds that have a Molecule Max Phase of 2, 3 or 4 – corresponding to evidence of safety in humans. You can do that by filtering via the filtering pane. Annotate their names and codes and the highest phase the compounds have reached. The results can be downloaded as csv file via the corresponding button.
- Repeat this process for all your prioritized targets.
- At the end of this step, a list of compounds that interact with the prioritized disease targets in step 4.2 will be generated. For each compound in the list, additional data like its phase of development, the target, and other information available in ChEMBL will have been collected to support the repurposing hypothesis.

Prioritization of selected compounds

The prioritization of compounds in the list may include those approved or at later stages of development but ideally focuses on those that have reached the market (with Molecule Max Phase=4). More information about approved compounds, or those in clinical phases, in ChEMBL can be found in the link: <https://chembl.gitbook.io/chembl-interface-documentation/frequently-asked-questions/drug-and-compound-questions>

1. Prioritization of compounds through physicochemical properties exploration

In this section of the workflow, prioritization of selected compounds is proposed on the basis of the analysis of known or predicted physicochemical properties. As an example, compounds may be prioritized by calculating or measuring their capabilities to cross biological barriers, like the blood-brain barrier for compounds targeting the Central Nervous System.

One way to assess the ability of the selected compounds (from the list assembled in step 4.3) to penetrate the central nervous system (CNS) is by looking at a molecule's lipophilicity, a physicochemical property often measured by LogP (otherwise known as the partition coefficient).

The partition coefficient is the ratio of the concentration of a compound in a hydrophobic environment vs. the concentration of that compound in a hydrophilic environment.³

³ ACD/Labs, LogP—Making Sense of the Value, https://www.acdlabs.com/wp-content/uploads/download/app/physchem/making_sense.pdf

Many databases will report the XlogP3 metric – this is simply a method of calculating the LogP of a molecule computationally (rather than experimentally) based on the individual contributions of its constituent atoms.

To explore the experimental or calculated LogP of the different compounds you can use the following databases:

- [PubChem](#): PubChem⁴ is a database that contains millions of chemical compounds, together with some annotated data including interactions with targets.
- [CompTox](#): The CompTox Chemicals Dashboard^{5,6,7} provides public access to chemical data. It is a widely used resource for chemistry, toxicity, and exposure information for over a million chemicals.

The workflow below is a suggested approach to assess blood-brain barrier penetration of your compounds using lipophilicity.

For each compound in the list generated in step 4.3:

- Go to PubChem, enter the name and/or ID of your compound.
- Find the “Chemical and Physical Properties” data.
- Find the XlogP (or XLogP3) value – record this for each compound.

For the list of compounds selected for the repurposed indication, the annotations at this stage should include their possible targets, their development phase, and their XlogP values.

When prioritizing compounds for repurposing, based on physicochemical properties like the partition coefficient, the following guidance could be used:^{8,9}

- A drug targeting the central nervous system (CNS) should ideally have a logP value around 2.2.
- For oral and intestinal absorption, the ideal value is 1.35–1.8.
- A drug intended for sub-lingual absorption should have a logP value >5.

2. Prioritization of compounds through safety exploration

In this section of the workflow, prioritization of selected compounds is proposed on the basis of known or predicted safety events associated with the selected compounds. There are postmarketing reports of safety events associated with approved compounds that are collected in Spontaneous Reporting Systems databases by regulatory agencies. It is important to consider

⁴ Research Boulevard Technologies (2021) *How to browse PubChem? How to find structural Analogs in PubChem?*, <https://www.youtube.com/watch?v=Jzqry-qoUUw>

⁵ <https://www.epa.gov/comptox-tools>

⁶ For more information you may visit: <https://www.epa.gov/comptox-tools/comptox-chemicals-dashboard-about>

⁷ Research Square (2017) *The CompTox Chemistry Dashboard: Helping researchers ID unknown contaminants*, https://www.youtube.com/watch?v=ObANKz_iVRl

⁸ The following reference provides more guidance on the interpretation of logP values: https://www.acdlabs.com/wp-content/uploads/download/app/physchem/making_sense.pdf

⁹ The paper gives a more detailed overview of the medicinal chemical properties of successful CNS drugs: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1201314/>

these safety events when studying a new compound for a repurposed indication, since patients of the new disease may have different safety thresholds (therapeutic margins) than other patients for which the repurposed compound was originally developed.

There is a need to understand which compounds are unsafe in your patient population, which may have negative interactions with other medications for the condition, and which may have safety effects that could be particularly problematic or widely accepted by the patient community.

For each compound in the prioritized list after step 4.4, you could extract:

- The main targets and off-targets reported (i.e. which proteins the compounds interact with)
- The main safety events reported once these compounds have been taken by patients. These safety events are taken from the Pharmacovigilance reports and recorded in different databases.

One way to analyze these data is through the software platform [ClarityVista](#) that has been developed by the REMEDI4ALL partner Chemotargets¹⁰.

ClarityVista integrates and curates data from many different databases across the drug discovery and development path, including:

- Safety pharmacology: how chemical and biologic compounds interact with targets (on targets for efficacy and off-targets for safety)
- Preclinical Toxicology; toxicology of compounds observed in *in vivo* experiments with animals
- Clinical Safety; safety events observed in Clinical Trials in humans for compounds in development
- Post-marketing Reports (Pharmacovigilance); reports of safety events included in databases like FAERS and VAERS (from the Food and Drug Administration, FDA, in the USA); Vigibase (from the World Health Organization (WHO)); and JADER (from the Japanese regulatory agency, PMDA).

The ClarityVista database can be explored via compounds, targets, indications, and safety events.

For each compound in the list compiled and prioritized at step 4.4, using ClarityVista would allow to explore their possible targets (under “safety pharmacology” in the platform), and their safety events (under “signal analysis” in the platform). ClarityVista classifies serious side effects using the following scale, in a decreasing order of severity:

- Signal
- Warning
- Reported

¹⁰ Learn more about Chemotargets here: <https://chemotargets.com/>

Investigation of these safety signals and off-target effects for the prioritized repurposing candidates allows to record the side effects that have an alert level of “Signal” for each of the compounds in the list. This level is associated with a statistically significant number of pharmacovigilance reports for a particular side effect.

Finally, having gathered all the information in steps 4.1 to 4.5, the repurposing compound candidates can be prioritized through the assessment of the different variables investigated, analyzing the following questions:

- What targets do the selected compounds interact with, and could this interaction have a potentially therapeutic effect for the selected indication?
- What are the physicochemical properties of the prioritized compounds? Could the compound plausibly be delivered to the tissue / organ where it should carry out its effect?
- What are the off-target effects of the prioritized compounds?
- What are the major safety concerns of the prioritized compounds?
- Are the prioritized compounds readily available on the market today, or are they nearing the market, for other indications than the selected repurposing indication?

5 Further applicable resources

REMEDI4ALL has developed the website <https://idrc-r4a.com/landing>, which explains the *in silico* drug repurposing process through text and video, and allows to access the REMEDI4ALL *in silico* drug repurposing catalogue at <https://idrc-r4a.com/>.

6 Annex

The file REMEDI4ALL Hackathon_full guide_v3.docx contains the guidelines used in the REMEDI4ALL 1st Hackathon, which include a practical exercise on the selection of compounds for drug discovery repurposing for different neurodevelopmental disorders indications.

